

3x3 Beginner

By Kunal Oak

Solving First Layer Corners

Hold corner on top of its correct spot and repeat this algorithm -

R U R' U'

until the corner is solved

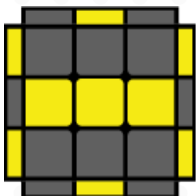
Solving Middle Layer Edges (First 2 Layers)

Left insert : **U' L' U L U F U' F'**

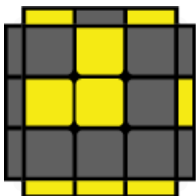
Right insert : **U R U' R' U' F' U F**

2 difficult cases covered in video tutorial

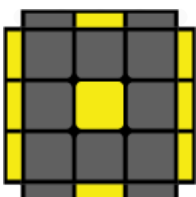
Orienting all 4 Yellow Edges (3 cases)



Line case : **F R U R' U' F'**



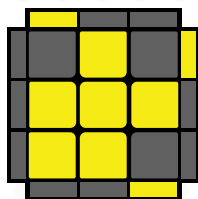
L case : **F R U R' U' F'**
(Reduce to Line case)



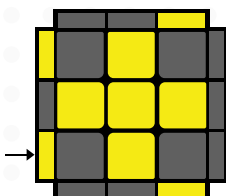
Dot case : **F R U R' U' F'**
(Reduce to L case)

Corner Orientation

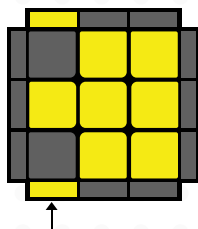
Corner Orientation Algorithm : **$R\ U\ R'\ U\ R\ U^2\ R'$**



One yellow corner oriented : Hold oriented corner in front left.



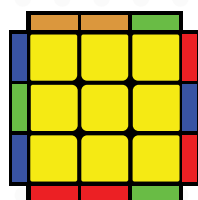
None yellow corner oriented : Front left corner should have yellow facing LEFT
(Reduces to case-1)



Two yellow corners oriented : Front left corner should have yellow facing FRONT
(Reduces to case-1)

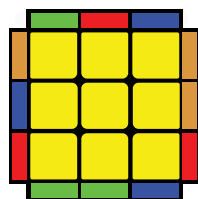
Corner Permutation

2 adjacent corners solved (headlights)



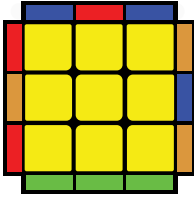
T-Perm: **$R\ U\ R'\ U'\ R'\ F\ R^2\ U'\ R'\ U'\ R\ U'\ F'$**
(Hold Solved Corners on Left)

No Corners Solved



T-perm: **$R\ U\ R'\ U'\ R'\ F\ R^2\ U'\ R'\ U'\ R\ U'\ F'$** (any angle)
(Reduces to case-1)

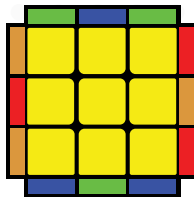
Edge permutation



1. 1 edge solved: (solved Edge in back)

Ua : $R U' R U R U R U' R' U' R^2$

(Either cube will be solved or you have to repeat the algorithm one more time)



2. None of the edges are solved (any angle)

Ua : $R U' R U R U R U' R' U' R^2$

(Reduces to case-1)

Optimizing the beginners method (first layer)

1. Directly solve the Cross Edges on to the white face instead of daisy method

2. Solving a Cross Edge i.e, Flipped
(Hold flipped edge on top right)

$R' U F' U'$



3. 3 Cases for first layer corner

a. White is facing right **R U R' U'**



b. White facing the top : **R U2 R' U' R U R' U'**

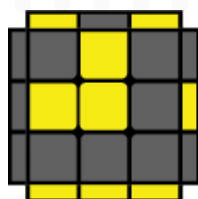


c. White facing front : **U R U' R'**



Second Layer

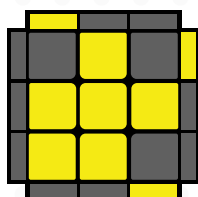
Flipped edge: **R2 U2 F R2 F' U2 R' U R'**



Third Layer

Optimizing the Edge Orientation

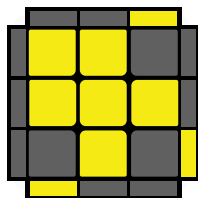
L Case : **F U R U' R' F'**



Optimizing Corner Orientation : 2 cases

One Oriented Corner

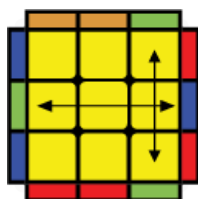
Case 1: Yellow facing front : **R U R' U R U2 R'**



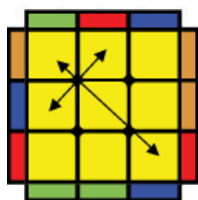
Case 2 : Yellow facing right : (Hold Oriented Corner in Back Left)

$R' U' R U' R' U^2 R$

Optimizing Corner permutation



1. T-perm : **$R U R' U' R' F R^2 U' R' U' R U R' F'$**

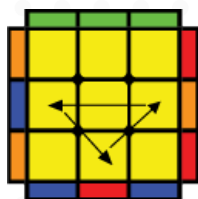


2. No pair of corners solved : instead of solving T-perm twice

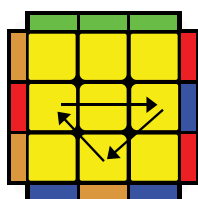
Y- perm: **$F R U' R' U' R U R' F' R U R' U' R' F R F'$**

Optimizing Edge permutation

2 cases



1. U_a : **$R U' R U R U R U' R' U' R^2$** (solved face in back)



2. U_b : **$R^2 U R U R' U' R' U' R' U R'$** (solved face in back)